

STEP 1 — CORE CRM FOUNDATION (Perfex-Based GHL Clone)

We will go **extremely detailed**.

Don't rush — this step is the **base of everything**.

STEP 1 GOAL

Build a **GHL-style CRM core inside Perfex** with:

- Lead Management (enhanced)
 - Pipeline (Opportunities)
 - Activity Timeline (GHL-style)
 - Conversation tracking (basic structure)
-

1.1 DATABASE DESIGN (VERY IMPORTANT)

You already have Perfex tables like:

- `tblleads`
- `tblclients`
- `tblcontacts`

We will **extend**, not break existing system.

1.1.1 Add Pipeline System

- ♦ **Create Table: `tbl_pipelines`**

```
CREATE TABLE tbl_pipelines (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(191) NOT NULL,  
  company_id INT DEFAULT 0,
```

```
is_default TINYINT(1) DEFAULT 0,  
created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

♦ Create Table: **tbl_pipeline_stages**

```
CREATE TABLE tbl_pipeline_stages (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  pipeline_id INT NOT NULL,  
  name VARCHAR(191),  
  stage_order INT DEFAULT 0,  
  color VARCHAR(20),  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

♦ Modify **tblleads**

Add fields:

```
ALTER TABLE tblleads  
ADD pipeline_id INT DEFAULT NULL,  
ADD stage_id INT DEFAULT NULL,  
ADD lead_value DECIMAL(15,2) DEFAULT 0,  
ADD last_activity DATETIME DEFAULT NULL;
```

Result:

You now have:

- Multiple pipelines
 - Drag-drop stages (Kanban ready)
 - GHL-style opportunity tracking
-

1.2 LEAD → OPPORTUNITY LOGIC

In GHL:

Lead = Contact
Opportunity = Deal inside pipeline

✓ Implementation Strategy

We will:

- Keep using `tblleads`
 - Treat it as BOTH:
 - Lead (raw)
 - Opportunity (when pipeline assigned)
-

🔄 Logic:

Condition	Meaning
<code>pipeline_id = NULL</code>	Normal lead
<code>pipeline_id != NULL</code>	Opportunity

✓ Add Helper Function

```
function is_opportunity($lead_id) {  
    $CI = &get_instance();  
    $CI->db->where('id', $lead_id);  
    $lead = $CI->db->get('tblleads')->row();  
  
    return !empty($lead->pipeline_id);  
}
```

🎯 1.3 KANBAN PIPELINE VIEW (LIKE GHL)

✓ Frontend Requirement

Create page:

`/admin/pipelines/kanban`

UI Structure:

[Stage 1]	[Stage 2]	[Stage 3]
Lead A	Lead C	Lead E
Lead B	Lead D	

✓ Backend API

Get Pipeline Data

```
public function get_pipeline_data($pipeline_id)
{
    $this->db->where('pipeline_id', $pipeline_id);
    $leads = $this->db->get('tblleads')->result_array();

    $grouped = [];

    foreach ($leads as $lead) {
        $grouped[$lead['stage_id']][] = $lead;
    }

    return $grouped;
}
```

✓ Drag & Drop Update

When user moves lead:

```
$.post('/admin/leads/update_stage', {
    lead_id: leadId,
    stage_id: newStageId
});
```

Controller:

```
public function update_stage()
{
    $lead_id = $this->input->post('lead_id');
    $stage_id = $this->input->post('stage_id');

    $this->db->where('id', $lead_id);
    $this->db->update('tblleads', [
        'stage_id' => $stage_id,
        'last_activity' => date('Y-m-d H:i:s')
    ]);

    // Trigger automation later (Step 2)
```

}

1.4 ACTIVITY TIMELINE (VERY IMPORTANT - GHL CORE)

This is **must-have** — GHL is strong because of timeline.

Create Table: **tbl_lead_activities**

```
CREATE TABLE tbl_lead_activities (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  lead_id INT NOT NULL,  
  type VARCHAR(50), -- call, sms, email, note, status_change  
  description TEXT,  
  created_by INT,  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

When to insert activity?

Trigger on:

Event	Type
Lead created	created
Stage changed	stage_change
Email sent	email
WhatsApp sent	whatsapp
Note added	note

Example Insert

```
function log_activity($lead_id, $type, $description)
```

```
{  
  $CI = &get_instance();  
  
  $CI->db->insert('tbl_lead_activities', [  
    'lead_id' => $lead_id,  
    'type' => $type,  
    'description' => $description,  
    'created_by' => get_staff_user_id()  
  ]);  
}
```

✓ Show Timeline UI

In Lead Profile Page:

[10:30 AM] Lead Created

[10:35 AM] WhatsApp Sent

[11:00 AM] Stage moved to Qualified

1.5 CONVERSATION SYSTEM (BASE STRUCTURE)

We will expand later, but create base now.

✓ Table: **tbl_conversations**

```
CREATE TABLE tbl_conversations (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  lead_id INT,  
  message TEXT,  
  direction ENUM('inbound','outbound'),  
  channel VARCHAR(50), -- whatsapp, email, sms  
  status VARCHAR(50),  
  created_at DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```

✓ Usage

- Store all communication

- Later connect:
 - WhatsApp API
 - Email API
-

1.6 HOOK SYSTEM (IMPORTANT FOR STEP 2)

We prepare for automation.

Create Hook Function

```
function trigger_event($event, $data = [])
{
    $payload = [
        'event' => $event,
        'data' => $data
    ];

    // Send to n8n later
    $ch = curl_init('https://your-n8n-url/webhook/'.$event);
    curl_setopt($ch, CURLOPT_POST, 1);
    curl_setopt($ch, CURLOPT_POSTFIELDS, json_encode($payload));
    curl_setopt($ch, CURLOPT_HTTPHEADER, ['Content-Type: application/json']);
    curl_exec($ch);
}
```

Use It

Example:

```
trigger_event('lead_created', $leadData);
trigger_event('stage_changed', $leadData);
```

1.7 TESTING CHECKLIST

Before moving to Step 2:

Must Work:

- Create pipeline
 - Add stages
 - Assign lead to pipeline
 - Drag & drop stages
 - Activity logs working
 - Timeline visible
 - Conversations table inserting manually
-

STEP 1 OUTPUT (WHAT YOU SHOULD HAVE)

After this step, your system should behave like:

-  Basic GHL CRM
-  Pipeline (Kanban)
-  Opportunity tracking
-  Activity timeline
-  Event trigger system ready